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# LEARNING[1,2]

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## THEORY

Books by Hull (74) and Guthrie (62) have brought their respective $S$-$R$[3] theories up to date. A brief article by Tolman (165) gives a précis of his most recent general formulation. MacCorquodale & Meehl (110) have presented an elaboration of an earlier paper (112) in which they formalized expectancy theory by a set of postulates. They include reinforcement in addition to frequency of presentations, $S_1 \rightarrow R_1 \rightarrow S_2$, as probably an important determiner of expectancies.[3] This is the only point in their discussion to which the present reviewers would definitely take exception.

Hull's last book (74) is of great significance and will long stand as a memorial to his creative genius. It presents in final form his basic postulates and corollaries. It also contains many ingenious, though often not very rigorous, applications of these postulates and corollaries, through the derivation of theorems, to most of the factual problems of animal learning. One feature of the system as now presented, in which we naturally concur, is his latest equation for Reaction Potential,

$$_sE_R = D \times V_1 \times K \times {}_sH_R$$

(where $D$ stands for drive strength, $V_1$ for stimulus-intensity dynamism, $K$ for incentive motivation, and ${}_sH_R$ for habit strength). This formula makes somewhat the same distinction between learning $({}_sH_R)$[3] and performance $({}_sE_R)$[3] that we would. Two basic differences between Hull's system and ours are, however, to be noted: (*a*) Hull conceives what is learned to be an ${}_sH_R$ whereas we conceive it to be an expectancy, a "readiness" for an $S_1$ to be followed by an $S_2$ provided a certain type of $R_1$ occurs; and (*b*) Hull still asserts the acquisition of an ${}_sH_R$ to be a function of frequency of reinforcement (though no longer of amount of reinforcement). We would assert the acquisi-

[1] This review is based on publications covering the period from about May 1, 1952 to May 1, 1953. The writers wish to express their gratitude to Mr. J. Trevor Peirce for bibliographical assistance, and to Miss Pauline Austin for help in reading the proofs.

[2] The following symbols appear in this chapter. When special meanings are needed additional symbols are added to the basic letters. The meanings intended are explained by the authors where these first appear. CR (conditioned response); CS (conditioned stimulus); $D$ (drive, drive strength); ${}_sE_R$ (excitatory strength); ${}_sH_R$ (habit strength); $I_R$ (reactive inhibition); ${}_sI_R$ (conditioned inhibition); $M$ (means); $pR$ (pointer-reading response); $R$ (response); $r_G$ (fractional anticipatory goal response); $rs$ (central neural surrogate response); $S$ (stimulus); $s_G$ (fractional goal stimulus); US (unconditioned stimulus).

[3] We have gratefully borrowed some features of their notation.



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tion of an expectancy to be primarily a function of frequency of past occurrences of $S_1 \rightarrow R_1 \rightarrow S_2$, whether or not $S_2$ has had positive, negative or zero incentive value. However, positive or negative incentive values acting as perceptual emphasizers undoubtedly speed up acquisition.

Gleitman *et al.* (53) have attacked Hull's doctrine of extinction. They point out, for example, that insofar as Hull now defines responses not as specific muscle contractions but as oriented adjustments to the environment, $I_R{}^2$ (conceived as attributable to the building up of specific fatigue-like products) loses much of its applicability. Further, insofar as extinction is also said to be due to the building up of ${}_SI_R{}^2$, progressive increase in the latter should eventually overtake the amount of ${}_SH_R$ and cause the response to disappear permanently. Gleitman *et al.* also point to the difficulties, as had Ritchie (134), of the concept of the "not-response."

Seward has continued his bridge between *S-R* theory and expectancy theory by his construct of $rs^2$ (cf. 143). The *rs* performs for Seward much the same function that $r_G{}^2$ and $s_G{}^2$ perform for Hull and Spence. The *rs* is an assumed neurological mechanism by which the significate and also its goal character, if it should have one, are brought back to the *S* at the choice point and made a determiner of *R* at that point. In two recent papers (144, 145), he has specifically applied this concept of *rs*'s to the acquisition of motives.

The new edition (62) of Guthrie's book presents again his doctrine of single-trial *S-R* contiguity learning. Guthrie also asserts once more his doctrine that the *R* which gets conditioned is the "movement" which removes the organism from the given S. So-called adjusted, purposive "acts" are for him purely fortuitous, happy (or perhaps unhappy) combinations of conditioned "movements." It would seem to the reviewers that nature must be even cleverer about providing the right environmental stimuli for combining movements into adjusted, directed acts than we would have supposed her capable of.

There have also appeared several discussions concerning the legitimate objectives of learning theory. Kendler (81) argues that a behavioral theory can as such merely seek for operationally specifiable intervening variables. Insofar as the theorist gives substantive meanings to these intervening variables (i.e., asserts hypothetical constructs) he is going in for reifications which are not legitimate since they have no truth value. These constructs may, however, be helpful to the particular theorist, given his private style of thinking. Ritchie (135) in an oblique and witty reply points out that, if geographers had held to this same "modern methodology," they would not have found it permissible to ask whether the earth were round or flat! But, if Kendler be right, they should have arrived at the same rules for navigation. Ritchie's moral appears to be that if one assumes, for example, the model properties of *S-R* connections one really will arrive at different predictions for behavior than if one assumes the model properties of expectancies. Davis (32), at odds with all theoretical constructs at the merely behavioral level, suggests that what we need is a "physical psychology" in which stimuli, responses,

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and intervening processes would all be defined by their physical and physicochemical characteristics. Kessen & Kimble (83) in criticizing Krech's neurological "dynamic systems" (90, 91) argue valiantly for the legitimacy of any sorts of theoretical constructs be they drawn by analogy from neurology, physics, sociology, aesthetics, or whatever.

Our classification of the empirical studies reflects the fact that our systematic approach leads us to employ four types of constructs in the analysis of stimulus-response associations, viz: discrimination, expectancy, drives and incentives, and performance. In line with this conceptual analysis we shall distribute our summaries of the empirical studies under the headings: (*a*) discrimination learning; (*b*) acquisition of expectancies in classical and instrumental conditioning; (*c*) performance in classical and instrumental conditioning, especially as determined by drive conditions; and, cutting across these categories, (*d*) verbal learning, and (*e*) motor learning.

## DISCRIMINATION LEARNING

The process of discrimination is central to an expectancy theory of learning. In such a theory, the organism is conceived to discriminate and categorize environmental stimuli as signs, means, and significates. Our analysis of discrimination learning will focus on two major problems: (*a*) conditions governing perceptual discrimination per se; (*b*) the ways in which the discriminated features of the environment come to function as signs.

### Conditions of Discrimination

When differential responses are systematically related to an orderly series of stimuli (such as black, gray, and white), a dimension of discrimination has been established. In experiments on discrimination learning, stimuli located at different points along such dimensions come to serve as differential signs to positive and/or negative significates. The more firmly the relevant dimensions of discrimination have been established, the more readily can expectancies be learned. A number of studies help to throw light on the process of perceptual learning in general as well as on the acquisition of specific dimensions of discrimination.

*Early learning.*—Hebb's (69) theory of perception, with its emphasis on the importance of early perceptual learning, has stimulated a number of empirical studies. Forgays & Forgays (47) and Hymovitch (76) studied the effects of varied infant environments on problem-solving by the adult rat. Both found that early opportunity for perceptual experience may have lasting beneficial effects. Partial support for this conclusion also comes from a study of Bingham & Griffiths (15).

The influence of early training on discrimination learning was studied by Siegel (149) who showed that ring doves reared under conditions of visual deprivation were significantly retarded in the later solution of a discrimination problem. According to Griffiths & Stringer (59), variations in the inten-

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sity of early stimulation were not, however, found to produce significant effects on adult learning in rats.

*Perceptual training experiments.*—Hypotheses about perceptual learning have also been tested in experiments investigating the effects of training on the discrimination of specific classes of stimuli. Lawrence, who had previously presented important evidence for the acquired distinctiveness of cues (98, 99), used a transfer situation to study the conditions governing the establishment of a new dimension of discrimination in rats (100). A difficult discrimination is established more effectively if it is preceded by training on an easy discrimination along the same dimension than if all training is given directly on the difficult stimulus pairs. A gradual change from an easy to a difficult discrimination is optimal for the establishment of the dimension. We would certainly agree with Lawrence's conclusion that increase in ${}_{S}H_{R}$ is not the only factor in discrimination learning, and that the animal must learn "to isolate functionally the relevant stimulus dimension from other background and irrelevant cues" (100, p. 516).

Several experiments have demonstrated the effects of training on perceptual discrimination in human subjects. Smith (152) was able to produce increases in the apparent size of geometric forms by giving his subjects controlled experience with shapes of large physical size. The effects of the experience were specific to the forms used in the training. Bevan & Zener (12), investigating the effects of practice on the perceptual thresholds for geometric designs, obtained general practice effects in recognition as well as increased sensitivity to the specific figures practiced. Atkinson & Ammons (8) found that latency of recognition decreases with practice on a given stimulus but failed to find evidence for a perceptual "learning to learn." Closely related in its implications to the studies just reviewed is an investigation of Forgays (46) which showed that (*a*) tachistoscopic recognition thresholds for words decrease with educational experience and (*b*) superior sensitivity to words presented to the right of fixation, as compared with words presented to the left of fixation, is similarly a function of educational level. These results, in agreement with previous findings from Hebb's laboratory (116, 125), point to the learned nature of highly specific visual discriminations.

The learning of a discrimination may often depend on the performance of an orientating response or "receptor-exposure act." The beneficial effects of training on discrimination may be due partly to the establishment of such orienting responses. Wyckoff (183) presents an elaborate theoretical discussion of the conditions governing the acquisition of observing responses in terms of reinforcement theory. He proposes that the discriminative stimuli come to function as secondary reinforcers for the observing response once the subject has learned to respond differentially to the discriminative stimuli. The author recognizes the danger of circularity in this argument and attempts to escape from it by a series of quantitative assumptions concerning the effects of primary and secondary reinforcement. The reviewers could not help but feel that his highly elaborate analysis represents a tenuous *ad hoc* use of the principle of secondary reinforcement.

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*Differential effectiveness of discrimination stimuli.*—In analyzing response to a given stimulus unit, we must take account of the organism's discrimination system. By discrimination system we refer to the organisms' propensities to categorize and order stimuli along dimensions of discrimination. The discrimination system is specified in terms of the psychophysical functions characteristic of the organism. Several studies have appeared during the year which contribute to the specification of the discrimination systems of several species of laboratory animals.

Amsel (5), using rats, studied the rate of acquisition of a brightness discrimination as a function of the duration of the discrimination stimuli which consisted of black and white runways preceding a choice point. He found speed of learning a direct function of stimulus duration. A previous study by Eninger (37), using auditory discrimination stimuli had, on the other hand, found the shorter durations more effective. Another study by Eninger (38) compared the effectiveness of compound discrimination stimuli involving both visual and auditory sense modalities with stimuli involving only the one or the other of these modalities. The compound stimuli were more effective and visual cues resulted in faster learning than did auditory ones. Greater effectiveness of visual stimulus components, as compared to auditory and tactual ones, was also reported by Fink & Patton (42). The relative effectiveness of color and form in the discrimination learning of monkeys was compared by Cole (30). In tests pitting color against form he found color to be the dominant cue; further tests showed, however, that form discriminations may be learned simultaneously with the dominant color discriminations. Further evidence for the simultaneous establishment of discrimination along two dimensions (size and brightness) is presented by Grice (57) in a study of visual discrimination learning in rats.

## The Discrimination of Signs

What are the ways in which the discriminated features of the environment come to function as signs? From the point of view of the present analysis the two stimuli in a discrimination setup come to function as differential signs, one pointing to a positive, the other to a negative significate. Hence, in an expectancy theory, the problem of the nature and temporal development of the differentiation between the positive and negative signs is an important issue.

*The problem of patterning.*—A significant contribution to this problem has been made by Bitterman and his associates in a series of papers which began some three years ago and of which several appeared during the current year (16, 161, 167, 182). The general thesis of these papers is that a distinction must be made between two nonadditive processes of perceptual organization in discrimination learning: (*a*) configurational learning which involves a "primitive, diffuse" perception of the situation, and (*b*) relational learning in which the organism comes to respond selectively to the abstracted separate sign features in the situation. This latter process of relational learning corresponds to Lashley's (96, 97) analysis of the discrimination process

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whereas configurational learning conforms to the description of Gulliksen & Wolfle (61). Using this two-factor theory, Bitterman and associates have been able to account for their own experimental findings which conform to neither Lashley's nor Spence's (153, 154) theories. Notable among the empirical findings was the speedier acquisition of a successive as compared with a simultaneous discrimination (179). This finding was not, however, confirmed by Spence (155). It may be mentioned in this connection that Loess & Duncan (103), comparing successive and simultaneous discrimination learning in humans, found (*a*) no difference between the two methods in learning an easy discrimination, and (*b*) simultaneous presentation superior in the learning of a difficult discrimination.

Spence (155) has criticized Bitterman's analysis, at the same time extending his own theoretical views so as to encompass the case of successive discrimination. Compounds such as "white-left" and "black-right" may acquire excitatory potential for the response of approaching. Such "transverse patterning" will, however, occur only under conditions which do not allow learning on the basis of simpler cue-position patterns. Bitterman (16), in a reply to Spence, summarizes the evidence for a two-factor theory and objects to the "theoretical and experimental constriction" of Spence's position.

The implications of Spence's theory of discrimination learning for transposition behavior were tested by Hunter (75) in a series of experiments on size discrimination in children. His results uniformly favor interpretation in terms of relational learning rather than reinforcement of absolute cue properties. On the other hand, a study by French (48), showing the dependence of discrimination learning in humans on the number of common stimulus elements receiving consistent reinforcement, is in agreement with an interpretation in terms of stimulus generalization. In this connection, mention should be made of an extension of Spence's analysis to multiple-choice learning in a paper by Cook (31).

*Discrimination reversal.*—Studies of discrimination reversal have frequently been viewed as "crucial experiments" for testing divergent interpretations of discrimination learning (as, for example, in the continuity-discontinuity controversy). A modified continuity position is presented by Gatling (50) in his investigation of the effects of repeated stimulus reversals. His results show progressive decreases in negative transfer effects as reversals are continued. He ascribes the increased efficiency of learning to the simultaneous operation of two factors: (*a*) a continuing build-up of habit strength which is transferred from reversal to reversal; and (*b*) the mastery of the reversal problem per se, i.e., essentially the acquisition of a reversal-learning set. Several studies of discrimination reversal in human subjects appeared during this year. In a series of three papers, Buss (22, 23, 24) addresses himself to the problem of "rigidity" in discrimination-reversal learning, i.e., to the conditions determining readiness to shift from old to new discriminations. He found (*a*) differential and partial reinforcement during orig-

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inal training more favorable to reversal than continuous reinforcement; (*b*) the ability to shift depends on the absolute values of the cues reinforced during training rather than on the relation among the cues. An interpretation of "rigidity" as learned behavior is also proposed by Schroder & Rotter (141) who show that the propensity to shift from one solution to another in a card-sorting problem depends on learned expectancies of alternative solutions. Results favoring the continuity hypothesis in human discrimination learning are reported by Walk (176). In two experiments reversal of the cues used in the preliminary training slowed down learning of a visual discrimination although the effects were of limited duration. Walk states that the positive stimulus appeared to have greater weight in determining discrimination than did the negative stimulus. It should be noted in this connection, however, that in a study of visual discrimination in rats Fitzwater (43) found preliminary training, which included experience with the positive and negative stimuli presented separately, more favorable than reinforced practice on the positive stimulus alone.

*Discrimination-learning sets.*—The continued confirmation of a series of related expectancies may give rise to generalized expectancies ("beliefs") embodying the common features of the specific expectancies. It is in terms such as these that we would conceptualize the learning sets demonstrated by Harlow (64). At the same time, we would point out that the phenomenon of learning sets underscores the inadequacy, or at least the limitations, of an analysis of discrimination learning based on the reinforcement of absolute cue values. Several experimental analyses of discrimination-learning sets appeared during the year. Harlow & Warren (66) report the establishment of learning sets in a series of 450 discrimination problems using planometric stimulus objects (the previously reported studies had used stereometric objects). Although learning sets were again demonstrated, the inter-problem improvement tended to be less than in the earlier studies, a result which is attributed to the greater similarity among the stimuli used in the new experiments. Riopelle (133) presents an experimental analysis of the acquisition of learning sets in monkeys for the solution of new and reversed discrimination problems. Initial disparities between the learning rates for the two types of problems gradually disappeared as the animals gained experience. Riopelle suggests that the acquisition of a learning set involves "transfer suppression," i.e., performances on successive problems in general, and reversal problems in particular, become increasingly independent of each other as the training continues. The course of the acquisition of discrimination-learning sets in chimpanzees and in children was studied by Hayes and associates (67). Their results point to qualitatively different stages in the acquisition of discrimination-learning sets: an early stage in which abstract principles are learned and a later stage characterized by the improvement of performance skills and work habits. The data fail to yield evidence for interspecies differences in the ability to acquire sets. In a further paper, Hayes

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*et al.* (68) investigate chimpanzees' ability to learn various numbers of discrimination problems simultaneously. A high degree of accuracy can be obtained with as many as 20 simultaneous problems.

We cannot leave the problem of discrimination learning without noting Harlow's (65) eloquent insistence that the motivation of his primate subjects in discrimination experiments is not to reduce visceral needs but to satisfy their curiosity and to manipulate environmental objects. Working in Harlow's laboratory, Butler (25) has presented convincing evidence for the strength and persistence of a visual-exploratory motive which is not derived from other biological drives.

## ACQUISITION OF EXPECTANCIES IN CLASSICAL AND INSTRUMENTAL CONDITIONING

Before turning to the actual studies a brief expansion of our theoretical analysis will be helpful. The basic independent variable governing the acquisition of an expectancy is, according to us, one or more occurrences of a "stimulus-response-stimulus" sequence. We shall symbolize such an event by $\{(S_1, M):R_1 \rightarrow S_2\}$. This is to be read: an initial stimulus $S_1$ is presented and with it an associated "behavioral support" (see Tolman, 164, p. 96f) or "means," $M$, and, given this $S_1$ and this $M$,[2] a response-act, $R_1$, occurs and results in the stimulus $S_2$. Moreover, since such a sequence usually occurs within some drive context and may take place more than once, and since $S_2$ may have either positive, negative, or zero incentive value, the complete symbol will be $D\Sigma\{(S_1, M):R_1 \rightarrow S_2^*\}$ where $D$ stands for an active drive or drives; $\Sigma$ for number of occurrences; and (*) for the positive, negative or zero incentive value of $S_2$. Further, the consequence of such $D\Sigma\{(S_1, M):R_1 \rightarrow S_2^*\}$ occurrences is conceived by us to be the acquisition of an expectancy, symbolized as $(\dot{S}_1:\dot{R}_1\dot{S}_2^*)$. Such an expectancy is to be read: if a "discriminated" stimulus of type $\dot{S}_1$ be present, and if the "discriminated" response-act of type $\dot{R}_1$ be made, a "discriminated" stimulus of type $\dot{S}_2$ with "discriminated" incentive value (*) will appear. Given such an expectancy, if there be a drive $D$ relative to which $S_2^*$ is either positive or negative, then upon the presentation of $S_1$, there will occur a disposition to "go to," "escape from," "avoid," or "merely wait for" $S_2^*$. And this disposition will betray itself by a specific performance or pointer reading response $pR$ (e.g., salivating, sweating, running, jumping, etc.) which is what the experimenter observes and measures.

In Table I are listed eight independent variables the consideration of which is suggested by the above analysis. These provide us with the categories under which we shall order the empirical studies.

*Type or intensity of drive, D.*—(*a*) Noxious light as $D$: Flynn & Jerome (45) were able to teach rats multiple-choice-box problems where $D$ was the drive to avoid strong light and $S_2^*$ a dark goal compartment. Caldwell & Womack (27) found also that albino mice could learn a maze to get out of light. (*b*) Intensity of relevant $D$: Teel (160) found, using rats, that rates for

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TABLE I

INDEPENDENT VARIABLES GOVERNING THE ACQUISITION OF EXPECTANCIES

1. Type or intensity of $D$.[3]
2. Type or intensity of $S_1$.
3. Features of $M$.
4. Quality and incentive value of $S_2$* (Type and amount of reinforcement).
5. Frequency and probability with which $S_2$* follows $R_1$ (Frequency and pattern of reinforcement).
6. Time delays between $R_1$ and $S_2$* (Delay of reinforcement).
7. Intertrial intervals.
8. Derived incentive value of $S_2$* (Secondary reinforcement).

both learning and extinction did not differ for deprivation periods of 1, 7, 15, and 22 hr. of hunger. Apparently strength of $D$ over a normal range does not affect acquisition though it may affect performance. (*c*) The effect of an irrelevant $D$ on conditioning: The effect of an irrelevant drive upon the establishment of an association is illustrated in an experiment by Bahrick (9). This investigator subjected rats to sensory preconditioning with light and buzzer under two degrees of food deprivation. He then conditioned the rats to buzzer as sign for shock. Subsequently, he trained them on the light as sign for shock. The previous pairing of light and buzzer under high deprivation led to more rapid learning to avoid the light. Spence & Taylor (158) and Spence & Farber (157), using eyelid conditioning, found that anxious subjects, as measured on items from the Minnesota Multiphasic Personality Inventory, condition more readily and extinguish more slowly than nonanxious subjects. In such a situation, anxiety may be considered an irrelevant drive. Bitterman & Holtzman (17), using the galvanic skin response (*GSR*) also found that anxious subjects selected by a number of clinical measures, conditioned more rapidly and extinguished more slowly than nonanxious subjects. (*d*) Effect of anxiety on trial and error: Farber & Spence (40) found with a stylus maze that anxious subjects did more poorly, especially on difficult choice points. Where difficult discriminations have to be made between $S_1$'s leading to different $S_2$*'s (blind ends versus true-path segments) anxiety may interfere with the building up of the appropriate expectancies or, as the authors suggest, tends to enhance indiscriminately all the response tendencies present.

*Type of intensity of initial stimulus, $S_1$.*—The readiness with which an expectancy is acquired may depend on the quality or intensity of the initial stimulus. It is to be noted, however, that we have already discussed most of the experimental evidence illustrating this relationship in the section above on conditions of discrimination.

*Features of behavior support, $M$.*—We shall consider here a group of experiments in which the environmental arrangements supporting the animal's response appear to be an important variable. (*a*) Time intervals in condition-

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ing (that is, environmental supports for waiting): White & Schlosberg (180) conditioned a *GSR* with light as $S_1$ and shock as $S_2$* with different intervals ($M$'s) requiring different $R_1$'s of waiting between $S_1$ and $S_2$*. They found that an interval of 0.5 sec. resulted in the fastest conditioning. They also note that the optimal time intervals for autonomic and skeletal conditioning are about the same. Fitzwater & Reisman (44) compared forward, backward, and simultaneous presentations of $S_1$ and $S_2$* in instrumental finger withdrawal. The forward setup where $S_1$ preceded $S_2$* by 0.5 sec. here also produced the clearest evidence of conditioning. Kaplan (80), using intermittent reinforcements, taught rats to push a lever where the resultant $S_2$* was the going off, for 1.1 min., of a noxious light. For different groups the intervals between reinforcements were 0.2, 0.5, 1.0, 2.0, 3.0, or 5 min. The longer the time intervals the slower the rate of responding. That is, the longer the average time between the $R_1$'s in any interval and the $S_2$* at the end of that interval the less readily was the expectancy formed. (*b*) Supports for space and response learning: Hill & Thune (70) ran two groups of rats on single-choice elevated mazes. Their findings, in contrast to the earlier one of Tolman, Ritchie & Kalish (166), was that a "turning" $R_1$ was acquired more readily than a "going towards a given place" $R_1$. However, their maze-path arrangements (i.e., $M$) were different from those of Tolman *et al.* and seem to us to have favored response learning since the response group always made a right-angle turn whereas the place group had to make both right angle and obtuse turns. Moreover, the goal place did not occupy as distinctive a position with respect to the source of light. (*c*) Introduction of new $M$: Page & Hall (126) extinguished a conditioned avoidance response by restraining the rats (introducing a new $M$) for a brief interval after the coming of the CS[2] in the former shock compartment for five trials, and then testing in a series of ordinary extinction trials. These rats extinguished more rapidly than a control group not given this first "restraint" experience.

*Quality and incentive value of the terminal stimulus, $S_2$* (Type and amount of reinforcement).*—Furchtgott & Rubin (49) ran rats, all 22 hours hungry, in a two-unit maze to different amounts of food: 2500 mg., 250 mg., 75 mg., 20 mg., and 0 mg. No differences in the learning curves were obtained save for the 0 mg. group which showed little manifest learning. We would suspect, however, that, if food had then been introduced, the 0 mg. group would have shown that they had achieved "latently" the same amount of actual learning. Bitterman, Reed & Krauskopf (18) compared the conditioning of *GSR* to light with a US (shock) of 3 sec. duration (i.e., strong negative reinforcement) and a US[2] (shock) of only 0.5 sec. duration (i.e., weak negative reinforcement). No differences were found. On the other hand, Spence (156) conditioned eyelid responses under two different strengths of US (air puff). On the next day half of each group of subjects was tested under strong and half under weak air puff. The number of CR's[2] was significantly more for the strong than for the weak US both in training and in the test. Again the question remains open as to whether the strength of the negative unconditioned stimulus affected learning or performance.

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A study by Seeman & Williams (142) is concerned with the "Hull-Leeper" problem of the conditions under which different expectancies can be attached to different drive states. The animal must learn to discriminate between two drives as signs to different $S_2$*'s. These experimenters taught rats, when hungry, to run around the left arm of a maze to the goal box and food and, when thirsty, to run around the right arm of the maze to the same goal box and water. For the experimental group the final common path and the goal box were black for food and white for water. For the control group, they were black for both food and water. The former group learned faster. The two more distinct $S_2$*'s correlated with the two different drives favored acquisition of the two expectancies.

In general it is our view that it is frequency of stimulus-response-stimulus occurrences and not amounts of reinforcement which primarily affect acquisition. We believe, in contrast to Hull, that learning may occur in the absence of reinforcement. This last hypothesis is of course tested in experiments on latent learning to which we turn next.

*Latent learning.*—Interest in this problem continues unabated. The question no longer is whether latent learning does or does not occur. Rather a number of specific experimental variables are being explored in the hope of specifying the conditions of latent learning more and more precisely. (*a*) Spatial position of irrelevant incentive: Thistlethwaite (163) and McAllister (108) both investigated the effect on latent learning of the spatial position of food (irrelevant incentive) relative to the water when preliminary training was under thirst, and water was on both sides. Both investigators found that, when the food was placed close to the water, the rats did not show latent learning, but when the food was at some distance prior to the water they did show it. (*b*) Degree of motivation for relevant incentive: Johnson (79), using a T-maze with food in one goal box but no water in either arm, ran rats with different degrees of thirst. He found that the greater this original thirst, the greater the number of trials to learn to run to the food side when the rats were made hungry. This finding suggests that the more thirsty animals were perceptually more set for water (even though it was not there) and hence less ready to perceive the food. (*c*) Drive discrimination: Thistlethwaite (163) also tested the effect of preceding practice in drive discrimination. Two thirds of his rats were given preliminary practice in going to the food end of a straightaway when hungry and to the water end when thirsty. It was only these animals which showed latent learning. These results confirm the earlier findings of Christie (29). All such findings suggest that when in latent learning experiments the animal is switched to a new drive he must discriminate this new drive as well as remember where the appropriate goal $S_2$* has been if he is to perform correctly. (*d*) General maze-wiseness versus learning of true path: Muenzinger & Conrad (119), after giving their rats one, two, or three days of nonrewarded training in a six-unit symmetrical maze, reversed the path and tested under reward. Prior to introduction of reward, errors had been dropping out progressively. However, when the true path was reversed there was negative transfer only for the three-day

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group. The authors therefore distinguished between a general maze-wiseness acquired in the first two days, and a specific latent learning of the true path acquired later. (*e*) Place of prefeeding: Meehl & MacCorquodale (113) ran hungry unrewarded rats in the Blodgett maze for five days. Then the animals were prefed in a maze-like section placed in front of the starting box. When put back in the maze they exhibited a further drop in errors. The authors ascribe the effects of this prefeeding
drive to the maze environment. This conditioned hunger drive adds to the total drive strength which multiplies with habit strength to produce ${}_{S}E_{R}$. It seems to the reviewers a surprising assumption that at 22½ hr. of deprivation one prefeeding experience would add a substantial increment to the effective drive. (*f*) Effectiveness of unrewarded preliminary exploration: Gilchrist (52) gave different amounts of preliminary exploration of a T-maze without food and then trained with food. The more the preliminary exploration the faster the subsequent learning. Control groups exposed to similar amounts of previous exploration but with food showed no faster subsequent learning. Again it appears that it is the frequency of presentation rather than presence of positive reinforcement as such which determines acquisition.

*Frequency and probability with which terminal stimulus, $S_2$*, follows the response $R_1$ (Frequency and pattern of reinforcement)*—(*a*) Frequency: The influence of frequency per se is by now so well established that there appears to have been only one study during the year dealing with this variable alone. Siegel & Foshee (150), training children to press a bar for a candy reward, found that greater frequency in the original training caused greater resistance to extinction. (*b*) Probability of reward (reinforcement): A number of studies have been concerned with speed of acquisition and with resistance to extinction as determined by various schedules of partial reinforcement. Grant & Schipper (55) used 25 per cent, 50 per cent, 75 per cent, and 100 per cent reinforcements in eyelid conditioning. In general the greater the percentage of reinforcement, the faster the conditioning. They also found that the 50 per cent and 75 per cent groups showed greater resistance to extinction. Lewis (101), using a gambling setup with boys, also found that 50 per cent and 60 per cent groups extinguished more slowly than a 100 per cent group. However, Rubin (140), using a panel-pushing response with rats with distributed practice (40 min. intertrial intervals), found greater resistance to extinction for the continuously reinforced group. Grant, Schipper & Ross (56), with eyelid conditioning, did not, however, find that intertrial spacing reduced resistance to extinction following partial reinforcement. They suggest that the presence of human verbalization may account for the difference between their results and those of Sheffield's (148). In two experiments Bitterman and associates compared 50 per cent reinforcement when presented randomly and when presented alternately. With both humans (106) and rats (168) the alternate group extinguished faster. It appears to the reviewers that the results of the partial reinforcement experiments continue to pose a serious problem for *S-R* reinforcement theory. It is equally true that ex-

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pectancy theory has not yet been elaborated to a point where it can give a consistent interpretation of this

the conclusion, however, that organisms can learn probabilities, i.e., that probability of reward is one of the environmental features about which expectancies are formed.

*Time delays between response, $R_1$, and terminal stimulus, $S_2$* (Delay of reinforcement).*—(*a*) Unfilled versus filled delay: Ferster (41) trained pigeons in a Skinner box with aperiodic partial reinforcement (average every 1 min.). He then introduced a 60-sec. delay (the box went dark) between successful peck and reinforcement; rate of pecking decreased. However, if the length of delay was increased gradually from 0 to 60 sec., the original pecking rate could be maintained. Each bird then filled in the delay with its own "superstitious" response. In another experiment birds learned to keep up the rate of pecking to one color of the target key which did not give food but which did (average every 1 min.) produce a second color of the target key in an arrangement by which this second color was followed regularly and automatically (after 1 min.) by food. The birds continued to peck but more slowly during this delay period. The pigeon apparently cannot learn to expect a delayed goal unless during the delay it can develop some sort of a bridging response ($R_1$) (superstitious or otherwise). (*b*) Long versus short delays: Logan (104) used a double-bar Skinner box. One bar delivered a pellet after 1 sec., the other after 5 sec. The short-delay bar was responded to faster and was preferred. Lambert & Solomon (94), after teaching rats to run a straight alley (with a number of jumps in it) to food, tested extinction by putting a block near the start, near the goal, or merely by removing food. Block near the start caused most rapid extinction, block near the goal the next most rapid, and no food produced the slowest extinction.

*Intertrial intervals.*—Stanley (159) extinguished a learned turning response in a single T-maze under massed (every 15 sec.) and under distributed (every 15 min.) extinction trials. There was faster extinction for the distributed trials. He also tested for extinction under satiation. The rats arrived equally soon at a 50 per cent choice whether their trials were massed or spaced. Baron (10) found with delayed eyelid conditioning no differences in original learning between a mean intertrial interval of 270 sec. and one of 120 sec. Vandermeer & Amsel (175) also found with eyelid conditioning no difference between spaced and massed trials. Teichner (162), teaching rats to depress a food tray in response to a signal, did find distributed trials better. Holland (72), teaching rats to choose between two levers, also found distributed trials better. As in other areas of investigation (see below under verbal and motor learning) the effects of distribution on acquisition and extinction seem to defy any simple generalization and to depend upon specific experimental conditions.

*Derived incentive value of terminal stimulus, $S_2$* (Secondary reinforcement).*—Although there is by now a considerable body of experimental evidence in support of so-called secondary reinforcement, the nature of the "secondary

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reinforcers" is still moot. Even from the point of view of a reinforcement theory the distinction between primary and secondary reinforcer has become difficult. The presence or absence of biological need reduction has failed as the criterion. From our point of view environmental features can acquire incentive value and thus serve as way stations (subgoals) to the attainment of already existing goals. To what extent such subgoals can become goals in their own right seems to us to be an unresolved experimental question.

McClelland & McGown (109) found that, if there was an irregular and unpredictable association between a circular goal alley and when and how the rats were fed in it, this goal alley acquired more "secondary reinforcement" (i.e., subgoal-incentive value) than it did for rats which had always been fed regularly at one point in it. Bugelski, Coyer & Rogers (21) found no evidence that pre-nonfeeding in a goal box (after learning) caused extinction of a simple runway response. This was a repeat, with additional controls, of the earlier Seward & Levy experiment (146) which had seemed to give evidence of secondary nonreinforcement.

## PERFORMANCE IN CLASSICAL AND INSTRUMENTAL CONDITIONING

If there is an expectancy ($\dot{S}_1 : \dot{R}_1 \dot{S}_2^*$), and if $S_2^*$ has either positive or negative incentive value, the organism will have a disposition to get to, or to get away from, $S_2^*$. The strength of the disposition is a joint function of the strength of the expectancy, the incentive value of $S_2^*$ and the drive condition. Specific performances serve as pointer-reading responses, $pR$'s, for inferring these dispositions to get to, or get away from, $S_2^*$. In short, we are maintaining the distinction between learning and performance suggested by Tolman (164) and Lashley (95), and more recently, reflected in the distinction between ${}_sH_R$ and ${}_sE_R$.

The studies now to be reviewed are ones which we interpret as throwing light on the ways in which expectancy, drive, and incentive value interact. Once an expectancy can be assumed to have been established, i.e., learning to be complete, the strength of the organism's dispositions to respond can be related to (*a*) type or intensity of $D$ at the time of testing or (*b*) difference between training and test conditions with respect to $S_1$, $M$, and $S_2^*$ (generalization with respect to sign, means, and significate). In this section we shall attempt an interpretation of the experimental data as regards these determinants of performance.

### Drive and Performance

*Intensity of drive.*—Hillman, Hunter & Kimble (71) trained rats under two degrees of thirst in a 14-unit maze. No differences in error curves appeared, but the stronger deprivation group ran faster. A third and a fourth group were started on weak and strong deprivation respectively and then each shifted to the other deprivation. The error curves were not different from those of the first two groups. But the shifts from weak to strong motiva-

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tion and vice versa caused respectively faster running and slower running. Deprivation affected performance but not acquisition. Variation in the strength of an irrelevant drive at the time of testing may also affect performance. Brandauer (19) trained rats in a Skinner-type box under one degree of hunger. He then divided his subjects into three groups and extinguished under three degrees of thirst. The greater the thirst the more persistent the bar pressing performance. These results supported Webb's earlier findings (178). Kendler *et al.* (82) taught rats to make a given turn in a T-maze to obtain food pellets. The rats were then subdivided into two groups, one 22 hr. thirsty and the other 3 hr. thirsty. When tested under these thirst conditions, significantly more of the 22 hr. thirsty group turned away from the food side. The greater the thirst the greater the negative incentive value of the expected food and the less the performance of going to it. (Kendler's own interpretation is in terms of the incompatibility of anticipatory goal responses).

Kohn (89) and Berkun, Kessen & Miller (11) have reported two important related studies in which plastic fistulas were sewn into rats' stomachs. Using rate of drinking milk (11) or rate of panel pushing for milk (89), they found that, in general, prefeeding with milk through the mouth was the most effective, injection of the same amount of milk directly into the stomach next most effective, while injection of the same volume of isotonic saline solution was least effective in reducing further drinking or panel pushing. In our interpretation of these results these three types of prefeeding reduced the drive to different degrees, and these resultant differences in drive intensity correspondingly influenced performance. (In a further study in this series Miller & Kessen (115) showed that milk injected into the stomach served as adequate reward for the learning of a T-maze even though it was less effective than milk taken by mouth.) Kaplan (80), in the study already mentioned above in which the rats were trained to turn off a noxious light, also tested the effects of lights of different intensities, 0 mL. to 2400 mL. Rate of responding increased rapidly from 0 mL. to about 500 mL. but decreased somewhat with lights from 500 mL. to 2400 mL. A too intense light caused less performance. Apparently too strong a drive stimulus had a depressant effect. Campbell & Kraeling (28) ran rats on an electrified straightaway in which the voltage in the start and middle sections was high and in the goal section lower. They found that the greater the drop from the same initial voltage the greater the running speed. Secondly, if the goal-box voltage was zero, all groups performed with the same high speed, irrespective of the starting-section voltage. Thirdly, a constant decrease in voltage between starting section and goal section had less effect when both start and goal voltages were high than when both were low. That is, the effect on performance of one and the same drop was greater as between two low voltages than as between two high voltages. (A Weber phenomenon?)

*Type of drive.*—Amsel & Roussel (6) taught hungry rats to run a straightaway to food in two successive goal boxes. After the running speeds had stabilized the rats were then "frustrated" in the first goal box on half their

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trials by not finding food in it. They definitely ran faster and showed lower latencies in leaving this first box on the frustration trials. The authors conclude that frustration in the first box was a new added drive which enhanced the performance of going to the second goal box. Zeeman & Radner (184) taught rats to cross the center of a tilting box to turn off light but could not teach them to cross to turn on light. It would seem plausible to us to assume that both expectancies concerning the appearance or nonappearance of the light were equally well established. But only the drive to get away from light produced performance.

It should be noted that some of the "latent-learning" studies summarized above might also have been considered with this group of studies. For in many of them the final test runs consisted in introducing a new $D$, appropriate to the previously irrelevant goal object, and when the new $D$ was introduced the incentive value of the irrelevant goal became positive and the performance of going to it appeared.

## Generalization

We may turn now to studies of generalization. In terms of our analysis such studies investigate the effect of differences between training and test conditions as regards $S_1$, $M$, and $S_2^*$.

*Generalization with respect to* $S_1$.—These are studies which have been treated conventionally under the heading of stimulus generalization. Brush *et al.* (20) trained pigeons in the Skinner box to peck at a lighted circle 1.4 cm. in diameter. The birds were then tested on circles ranging from 2.6 to 0.2 cm. in diameter. A bell-shaped generalization curve (with the peak at the training size) was obtained. There was somewhat more generalization to the larger sizes. Kling (88) ran rats in a Grice alley to white discs. Each group was trained on successive trials to two different areas (both positive). Subgroups were extinguished on a single large area or on a single small area. The to-be-expected generalizations were found. Rosenbaum (138) studied generalization along the dimension of height under different strengths of an irrelevant drive (namely, anxiety). Training human subjects to make a voluntary movement of a bar to a visual rectangle, he induced different degrees of "anxiety" by giving strong shock, weak shock, and buzzer to different groups, if the responses were not quick enough. He tested on rectangles of other heights and the to-be-expected generalizations for response frequencies and response amplitudes were obtained. Strong anxiety (strong shock) produced the most generalization. Different results were however, obtained for latencies. The more different the test rectangle, the shorter the latency. This last finding is puzzling. May it be that responses to the remote test stimuli were made only when the subjects did not take time to make the required discrimination? Kimble & Kendall (86) studied, with rats, the effects of changes in $S_1$ in an extinction setting. They extinguished a "wheel-turning" response which turned off shock and which had been conditioned to a strong light as the sign for coming shock. They found that, if they began the extinction with a weak

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light, increasing it by gradual steps, none of which, when introduced, produced much avoidance, extinction was more rapid than with the use of the original stimulus at full strength. This was conceived as a validation of Guthrie's (62) "toleration method" for extinction.

*Generalization wlh respect to M.*—There have been almost no experiments in this area. Studies falling here would be ones testing new maze routes (short cuts, *Umwege*), or new time-intervals, or the like, to discover whether and how the disposition to respond supported by the original $M$ will generalize to a new $M$ or new $M$'s. One recent study which might, however, very tentatively be classed here would be that of Notterman, Schoenfeld & Bersh (124). They extinguished a conditioned heart-rate response in human subjects by three methods: Group I had no instruction; Group II the instruction "now no shock will follow"; Group III the instruction "tap electric key and no shock will follow." Group III extinguished most readily, Group II next and Group I not at all during the 11 extinction trials. The provision, by means of the different instructions for the different groups, of different new $M$'s during the waiting period caused different degrees of generalization.

*Generalization with respect to* $S_2$*.*—There seem to have been no studies during the year which fall under this heading. Experiments concerning this problem would be ones which after acquisition varied the nature or amount of $S_2$* and related the subsequent dispositions to respond (performance) to such variations. Such data could also be used to gauge the relative incentive values of different $S_2$*'s. The extinction situation represents the extreme case in which the new $S_2$* has no incentive value and in this case the generalized readiness to respond is always lost in time.

## Performance in the Presence of Alternative Expectancies

Several expectancies can be established in the same situational setting. So far we have discussed the relation of performance to expectancy as governed by drive and conditions of generalization. Several experiments suggest, however, that where alternate expectancies are built up the correlated dispositions to respond are affected by additional determinants.

*Fixation.*—We conceive fixation as the situation in which performance according to an early acquired expectancy interferes with performance according to a subsequently acquired expectancy. Wilcoxon (181) used special reinforcement schedules in the Maier "frustration" setup (111) to induce a strong expectancy highly effective in performance. His findings suggest that fixation is to be explained not as "frustration-instigated" behavior but as due to the dominance of the response disposition correlated with the original expectancy. It would appear that such factors as the special reward schedules and primacy of acquisition lead to the development of strong and resistant response dispositions which later interfere with the appearance of the new more appropriate response disposition, even though there was evidence in terms of "lookings" and "hesitations" that the corresponding new espectancies had actually been acquired.

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*Regression.*—Aebli (4) trained rats to go to their preferred turn or to their nonpreferred side. He then shocked them just prior to choice. Those trained on the preferred side continued to choose that side; those trained on the opposite side "regressed" to the preferred side. He concludes that regression is to an unlearned preference.

*Alternation.*—Where two different expectancies for which the $S_2$*'s have equal incentive value are built up, as, for example, in a T-maze where both arms lead to food, animals show a tendency to alternate between sides. The following investigations are concerned with discovery of the conditions which favor or reduce such alternations. Riley & Shapiro (132) tested in a T-maze the effect of light versus heavy weights on doors and also the effect of massed versus distributed trials. Massed trials clearly produced more alternation. Heavy weights gave only slight evidence of producing more alternation. Rothkoff & Zeaman (139) tested amount of alternation in a simple T, as affected by preceding forced trials to one side, by changes of intramaze cues and by changes of extramaze cues. They found a tendency to go to a new "place" as well as to make a new "turn." They say ". . . the data did not do violence to a composite 'tired stimuli and tired response' type of theory." Montgomery (118), continuing previous studies, used the T-maze with opposite starting stems, and found that the rats were alternating with respect to goal places but not with respect to turns. Goodman, Moyer & Bunch (54) measured alternations in a simple T-maze filled with water in a nonhungry group of rats, a group subjected to shock, and a group subjected to airblast. No differences in tendencies to alternate appeared. We conclude that alternation is an expression of the alternate "fatiguing," not of muscles ($I_R$) but of response dispositions for, say, going to light versus going to dark or for making right-hand turns versus making left-hand turns.

## Pointer-Reading Responses

Percentage correct responses, running times, rates of lever pressing, amplitudes, latencies, trials to extinction, etc., have all been used as $pR$'s[2] and have been assumed to be highly correlated. But many experiments make this assumption of high intercorrelations doubtful, and this lack of consistency among measures continues to pose a serious problem for any system (ours included) which attempts to use them interchangeably as dependent variables. Two recent experiments illustrate this lack of consistency among measures. Hall & Kobrick (63) measured the strength of a runway response in terms of latency, running time, and resistance to extinction. Latency and running were significantly but not highly correlated, but neither of them was correlated with resistance to extinction. Logan (105) reanalyzed the data of his experiment mentioned above (104) in which he used a double-bar Skinner box. The two bars had different delays of reward (1 sec. and 5 sec.). He had three measures: preferential latency for the short-delay bar, per cent of choice of the short-delay bar, and the difference between the response speeds to the two bars presented singly. He found good correlation between differ-

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ential response speeds and preferential latency but no significant relationship between either preferential latency or differential response speeds and per cent choice of short-delay bar.

## VERBAL LEARNING

The theoretical framework for most studies of verbal learning has remained associationistic. The conditions governing the formation of associations among the members of a verbal series continue to claim experimental attention. From the general point of view represented in this review, such studies serve to specify the conditions under which exposure to conjunctions of verbal stimuli serves to establish expectancies concerning such conjunctions.

*Meaning and familiarity.*—The related variables of meaningfulness and familiarity are stressed in several publications. Noble (121) has presented a theoretical analysis of verbal meaning in *S-R* terms and devised an index of meaning based on the number of different associations elicited by a given word. Using this index, Noble (122) then investigated the role of stimulus meaning in serial verbal learning. A number of performance criteria (pointer-reading responses) showed a direct relationship between degree of stimulus meaning and learning proficiency. The finding of Kurtz & Hovland (92) that verbalization during observation of stimulus objects increases accuracy of recognition and recall may be mentioned in this connection. The verbalization may be assumed to increase the richness of the associative context aroused by the stimulus objects.

Variations in meaning and familiarity produce parallel effects. Hovland & Kurtz (73) report that familiarization with nonsense syllables prior to learning speeds up acquisition. Such familiarization reduces the number of trials to mastery by a more or less constant amount for lists of different lengths. The results are in conformity with Robinson's Law of Acquaintance. In terms of information theory, familiarization reduces the amount of information conveyed by each syllable in the list.

Information theory is also applied by Aborn & Rubenstein (2), who investigated the influence of previously learned rules of organization on the immediate recall of nonsense syllables. The rules were such as to produce systematic variation in the amount of information per syllable. The lower the rate of information the more syllables were recalled.

The effects of previously learned associations on recall were demonstrated by Jenkins & Russell (78). Stimulus and response terms from the Kent-Rosanoff list were scattered at random through a list of words. Recall tests showed a strong tendency for the Kent-Rosanoff pairs to be recalled together. In the recall, the *S-R* sequence of the Kent-Rosenoff test was also maintained.

*Conditions of practice and testing.*—Under this heading, we shall make brief reference to a somewhat heterogeneous group of experiments which have varied conventional conditions of practice or testing or both in investigations

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of serial association. Meyer & Miles (114) investigated changes in the process of rote learning which result from continued practice (20 successive lists). They were unable to find systematic changes in the mathematical properties of learning-curves at different stages of practice, and there is scant evidence for qualitative changes in the learning process.

An interesting variation in conventional procedures of serial learning is introduced by Deese & Kresse (33) in their study of the types of errors which account for the classical bow-shaped serial position curve. To gauge fully the response dispositions at different points in the series, subjects were given unlimited amounts of time for anticipation of each syllable. Two types of errors summate to form the serial position curve: a distribution of intralist intrusions symmetrical around the middle of the list, and failure-of-response errors which rise to an asymptote in the second half of the list.

Turning to conditions of testing, we note the failure of Rockway & Duncan (136) to find any evidence for pre-recall warming up in retention tests for serially learned verbal materials. They made every effort to duplicate the procedures of earlier investigators but could not reproduce the results. Another "now-you-see-it-now-you-don't" phenomenon! Also on the negative side are the results of Grings (60) who attempted an experimental test of the "extinction" theory of verbal forgetting proposed earlier by Underwood (169). Varying amounts of distraction introduced during the retention test failed to produce "disinhibition," i.e., did not lead to recovery of presumably extinguished verbal responses.

*Distribution of practice and reminiscence.*—The phenomena of distribution and reminiscence appear in both motor and verbal learning. There are important differences between the two areas, however, as regards both experimental facts and theoretical interpretations: (*a*) the range of conditions over which distribution makes a significant difference appears to be much more circumscribed in verbal than in motor learning; (*b*) the theoretical construct of reactive inhibition which has carried the main burden of theoretical interpretation of distribution effects in motor learning can be applied to verbal learning phenomena only with considerable difficulty. Over the last two years or so, Underwood has been conducting a systematic investigation of the effects of distribution in verbal learning. Several of these studies (170 to 174) appeared during the current year. His results emphasize the dependence of distribution effects on very specific experimental conditions and the lack of a general theory under which the experimental findings can be subsumed.

Among Underwood's findings we may briefly mention the following: (*a*) the activity used to fill the rest intervals may significantly influence the effects of distribution; thus the conventional activity of color naming induces a responding set which raises the frequency of errors as well as the frequency of correct responses (170); (*b*) whereas distribution has a beneficial effect on the learning of serial nonsense lists, no such influence is found with paired nonsense syllables (171, 172); (*c*) similarly, with meaningful materials (ad-

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jectives) serial learning is improved by distribution whereas paired associate learning is not (173, 174). It is important to note that careful analysis of the wide variety of learning and retention data obtained by Underwood fails to lend consistent support to any one theory of distribution, such as the reactive-inhibition theory or the differential-forgetting theory. Gibson's generalization hypothesis (51) also fails to handle some of the major findings of these studies. The theoretical analysis of distribution of practice in verbal learning is still a wide open problem, as was also found to be true for classical and instrumental conditioning.

The picture presented by recent studies of reminiscence in verbal learning is equally inconclusive. Using serial lists of nonsense syllables Archer (7) failed to find any evidence for reminiscence; variations in the association value of the syllables, the length of the rest interval, and in the rest-interval activity did not yield any reminiscence effects. The results flatly contradict earlier studies. Riley (131), on the other hand, obtained evidence for reminiscence in paired-associate learning with nonsense syllables when competing responses are attached to the stimulus terms. The results are at variance with a differential-forgetting theory and are consistent with a work-decrement hypothesis, although the exact nature and conditions of such decrements in verbal learning remain uncertain. In accordance with earlier studies, no reminiscence effects were found by Noble (122) in the serial learning of meaningful words. The description of reminiscence as a "now-you-see-it-now-you-don't" phenomenon (26) appears to be as apt as ever.

*The law of effect in verbal learning.*—Much of Thorndike's experimental evidence for the law of effect was obtained in verbal learning situations. Implications of the reinforcement hypothesis continue to be tested in such situations. Thus the problem of "learning without awareness" has been reopened. Thorndike had argued that such learning would be evidence for the automatic effects of reinforcement. Evidence for discrimination learning without awareness of the principle of solution is reported by Walk (176). Postman & Jarrett (129) found only scant evidence for learning without awareness in the acquisition of an associative response tendency. Their results do not, however, support invocation of a principle of insight. Skill in the application of a verbalized principle showed a gradual increase.

The spread of effect, originally offered as critical evidence for the law of effect, continues to be reanalyzed. Sheffield & Jenkins (147) offer new evidence in support of their argument that the guessing-sequence hypothesis is adequate to account for the spread phenomenon. In the absence of reward, duplication of other features of Thorndike's situation (e.g., fixed order of stimuli, set for recall of word-number connections) yields levels of response repetition equal to, or exceeding those obtained by Thorndike.

The role of anxiety in serial rote learning is analyzed by Montague (117) from the point of view of reinforcement theory (cf. similar studies by Spence and associates (40, 157). Anxiety, as a drive, is assumed to multiply with habit strength to determine excitatory potential of correct as well as incor-

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rect responses. Anxiety should aid performance if correct response tendencies predominate and should hinder performance in the presence of incorrect competing tendencies (which are enhanced by the drive along with the correct ones). This implication was verified in a design employing nonsense-syllable lists of varying difficulty. Anxious subjects performed better on the easy lists and were surpassed by the nonanxious subjects on the difficult lists involving many incorrect response tendencies.

*Failure and related motivational factors.*—Verbal materials have frequently been used in the investigation of motivational factors in learning and retention, particularly to test the concept of repression or hypotheses stemming from it. Experimentally induced failure is one of the techniques used to create motivational conditions appropriate to the testing of such hypotheses. Several studies of this type appeared during this year. Eriksen (39) compared memory for completed and incompleted tasks by ego-involved and control subjects. Ego-involved subjects (*a*) recall relatively more completed tasks and (*b*) show greater variability than do control subjects. Eriksen interprets his data as demonstrating the importance of ego-defense in memory. Aborn (1) compared the relative effects of experimentally induced failure on materials learned intentionally and incidentally. Ego-threat had significant effects on memory for the incidentally learned materials only. In the case of incidental memory, such factors as rehearsal and overlearning presumably do not counterbalance the motivational effects. Murdock (120) investigated the effects of both failure and retroactive inhibition on mediated generalization. Both types of treatment served to reduce the amount of mediated generalization: failure by leading to the "suppression," and retroactive inhibition by producing extinction, of the mediating verbal responses.

An implication of repression theory was tested by Laffal (93) who compared words which produce association disturbances and words which do not with respect to speed of learning and retention. He found (*a*) slower acquisition of the disturbing words, and (*b*) no differences in retention for the two types of words. These contradictory findings lend only indifferent support to a repression hypothesis.

We feel that the effects of failure, stress, and related conditions on learning and retention can for the present be adequately and parsimoniously handled by the more mundane concepts of current learning theories. The invocation of ego defense or repression seems to put too heavy a conceptual burden on rather slender empirical facts.

*Social attitudes.*—The danger of erecting precarious theoretical superstructures on limited and uncertain data seems to us illustrated in a study of Rokeach (137) concerned with attitude as a determinant of distortion in recall. Low- and high-prejudiced subjects were shown photographs of whites and Negroes with fictitious names attached to them. In a recall test, the high-prejudiced group recalled certain names (Gray, Greene, Lynch) more often than the low-prejudiced group to both Negro and white faces, whereas the low-prejudiced group was partial to the name Jones. To explain these

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rather puzzling results, the author invokes the following concepts: Freud's theory of slips, a continuum of cognitive reorganization ranging from assimilation to various degrees of compromise and contrast, tolerance af ambiguity, tolerance of incongruity, closure! The ratio of concepts to data appears somewhat high.

Doob (34) investigated the effects of attitude on incidental memory for controversial items and found quantity of recall to be associated with intensity of attitude. In the same investigation, the effects of serial position on items evoking different attitudes were studied. In the case of items evoking strong drives, primacy increases amount of recall; in the case of items evoking weak drives, recency is more effective, probably by reducing the amount of retroactive inhibition to which the items are subjected. The picture presented is highly complex but we note with interest the attempt to apply systematic learning principles to memory data obtained under naturalistic conditions.

Perlmutter & de Montmollin (128) compared the performance of groups and individuals in a rote-learning task. Groups were found to be superior to average individual learners in total scores, but the best individual performers surpassed the group. The learning performance of individuals was improved by prior participation in group work. In a further study, Perlmutter (127), using Bartlett's method of repeated reproduction, compared the products of individual and group memory. He concludes that some of the group products are not derivable from the individuals' products. The reviewers express their bewilderment at the implied resurrection of the unlamented late group mind!

## MOTOR LEARNING

For an expectancy theory, specific motor acts serve as "pointer readings" for inferring the organism's cognitive dispositions. Variations in motor behavior would, therefore, be treated primarily under the general heading of performance. However, expectancy theory has not developed a systematic description of motor learning. For an *S-R* reinforcement theory, on the other hand, motor learning provides important opportunities for testing hypotheses about both learning and performance. Much of the current experimental work in motor learning centers around the implications of Hull's concepts of reactive inhibition ($I_R$) and conditioned inhibition ($_sI_R$). The specific experimental designs are concerned primarily with the effects of post-practice rest intervals on the acquisition and performance of motor responses. The inhibition built up during practice is assumed to dissipate during the rest. Much effort is now being devoted to obtaining as precise information as possible concerning the conditions under which inhibition grows and dissipates and the quantitative laws of these phenomena. Without reviewing these studies in detail, we shall consider some of the main experimental problems and conclusions.

*Distribution of practice.*—Several studies have compared acquisition and

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performance of motor skills under distributed and massed practice. In general, spacing has a beneficial effect on acquisition and performance. The effects appear, however, to be transient. Thus Adams (3) reports that both his distributed and massed group were approaching the same asymptote of performance on the pursuit rotor. Using three different psychomotor tests, Reynolds & Bilodeau (130) found that distributed practice during acquisition resulted in superior performance on all tests. On tests of retention 10 min. and 24 hr. later the two groups become increasingly alike. Bilodeau (14) observed the effects of massing and distribution over an extended period of practice and found that performance depended more on the number of days of prior practice than on conditions of distribution. Kimble & Shatel (87) present data supporting Kimble's previous analysis (84) of inhibitory processes in motor learning. According to this view, reactive inhibition decreases with continuing practice since with declining motivation the subject will allow less and less $I_R$ to accumulate before he rests. Conditioned inhibition, on the other hand, is assumed to increase with practice in a negatively accelerated fashion. A review of the evidence supporting Kimble's theory is presented by Wasserman (177).

*Reminiscence.*—The reminiscence experiment is, of course, a special case of distribution of practice. The theoretical analysis of reminiscence effects in motor learning follows the same general line as the analysis of distribution effects: virtually all the investigators take their lead from the concept of reactive inhibition. A number of parametric studies has appeared, relating postrest gain to such variables as amount of prerest practice, duration of the rest period, and effortfulness of the task (e.g., 13, 36, 123). It would be difficult to summarize the experimental findings in terms of a few generalizations, beyond the fact that reminiscence does occur following massed practice. The reviewers are impressed with the dependence of the functional relationships on the specific experimental conditions.

Important questions concerning the interpretation of reactive inhibition are raised by a series of studies demonstrating reminiscence in bilateral transfer. Irion & Gustafson (77) found significant postrest gains in pursuit-rotor performance even though subjects shifted from right to left hand following the rest intervals. Similar results are reported by Kimble (85) whose subjects shifted from nonpreferred to preferred hand following the rest. Findings such as these throw doubt on a conception of reactive inhibition as localized in specific effector organs. Stimulated by this problem, Grice & Reynolds (58) compared the effects of varying amounts of rest on ipsilateral and bilateral transfer reminiscence. For both conditions, postrest gains were an increasing function of the amount of rest. Although the ipsilateral group showed larger gains than the bilateral group the data suggests that the general shape of the function is the same. Grice & Reynolds ascribe the postrest gains in bilateral transfer to "temporary inhibitory factors either associated with generalized postural adjustment or of a perceptual or central character." Such an interpretation seems to broaden the connotations of

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reactive inhibition very considerably. This general conclusion is in harmony with that reached above in connection with the experiments on alternation behavior in rats.

Rest intervals usually result in warm-up decrements which may be attributable to a loss of set or postural adjustment. Hence postrest performance may be improved by warming-up activities during the rest interval. Silver (151) has systematically explored the effects of varying amounts and distribution of warm-up activity during a rest interval. Warm-up activity clearly improved postrest performance provided the amount of such activity did not itself produce too high a degree of inhibition. Distributed warm-up activity is, therefore, more beneficial than massed warm-up activity.

*Transfer.*—Motor learning is convenient for the study of transfer processes because specific stimulus and response elements can often be identified precisely. Duncan (35) investigated transfer between two motor tasks as determined by two variables, degree of learning of first task, and intertask similarity. Degree of positive transfer was a direct function of both these variables. There was no evidence for interference between tasks. Both response generalization and "learning how to learn" may account for the uniformly positive transfer effects. Evidence for negative transfer effects appears, however, in studies of retroactive inhibition in motor learning. Lewis, Smith, & McAllister (102) studied retroactive inhibition in performance on a two-hand coordinator when original and interpolated tasks required a reversal of movements. They found amount of retroactive inhibition to vary with the degree of interpolated learning but, contrary to their expectations, differential amounts of retroactive facilitation were not developed. Using the same apparatus, McAllister (107) studied the effect of different degrees of original learning on retroactive interference. Interference increases as a function of original learning level, reaches a maximum and then appears to decline with overlearning of the original task (this finding is similar to results obtained with verbal materials). Relearning data showed the interference effects to be relatively enduring.

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